

Professional Portfolio

Personal Information

Name: Nathaniel Edem Aguze

Title: Computer Science Student and Statistician

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Location: University of Ghana, Legon



Education

Bachelor of Science in Computer Science

University of Ghana, Legon

Expected Graduation: [2027]

Relevant Coursework: Software Engineering (DCIT 208), Data Structures and Algorithms, System and Data Modelling, Probability and Statistics.

Specialization in Statistics

Focused on: Statistical modelling, time series analysis, and probability theory, including applications of least squares methods, seasonal indices, and the strong law of large numbers.

Skills

Technical Skills

- **Programming Languages:** Javascript, Java
- **Tools and Frameworks:** Selenium, WebStorm, Appium, Jira for QA testing; LaTeX for document preparation; Draw.io/Lucid chart for diagrams (ERD, UML, DFD).
- **Statistical Methods:** Linear regression (covariance matrix, Residual Sum of Squares), time series forecasting (multiplicative seasonal indices, ratio-to-trend), probability distributions (binomial, central limit theorem).
- **Software Development:** Mobile app testing (functional, usability, performance on iOS/Android), algorithm design (Dijkstra's, Vogel Approximation, Northwest Corner, Critical Path Method), searching/sorting (Merge Sort, Divide and Conquer, Greedy, Dynamic Programming).
- **Data Modelling:** UML class diagrams, sequence diagrams, entity-relationship diagrams, data-flow diagrams.

- **Other:** Vector algebra, projections, centroids; UI/UX design considerations (e.g., colour-blind accessible status indicators).

Soft Skills

- Strong analytical skills for identifying edge-case issues and optimizing solutions.
- Collaborative team player, experienced in group projects (e.g., team "Five9ine").
- Attention to detail and passion for delivering high-quality, user-centric applications.
- Ability to communicate complex technical concepts clearly.

Professional Experience

QA Tester (Mobile App Development)

Specialized in ensuring seamless user experiences through rigorous testing.

- Conducted functional, usability, and performance testing on iOS and Android platforms.
- Utilized tools like Selenium, Appium, and Jira to identify, track, and resolve defects.
- Collaborated with developers and designers to align on project goals and uncover edge-case issues.

Projects

University of Ghana Campus Route Finder (Java Application)

Developed an algorithm and application to find optimal routes on campus, considering shortest distance, travel time, and landmarks (e.g., Main Gate, Balme Library, Great Hall). *Techniques Used:* Dijkstra's algorithm, Vogel Approximation Method, Northwest Corner Method, Critical Path Method.

Features: Incorporated searching and sorting algorithms (e.g., Merge Sort) to provide at least three route options; user input for landmarks like "Bank" to generate efficient paths.

System Requirements Specification (SRS) for DCIT 208 Course Project

Led team "Five9ine" in creating a comprehensive SRS document.

Content: Included sections on Functional Overview, User Stories, Non-Functional Requirements.

Design: Created colourful ER diagrams and flowcharts using Draw.io (blue for entities, green for processes), exported at 300 DPI.

Deliverables: Prepared zipped deliverables for submission.

System & Data Modelling Assignment (E-Commerce System)

Created UML class diagrams, architecture diagrams, sequence diagrams, data models, and process models.

References: Kendall, Pressman, and Sommerville.

Details: Explained Data-Flow Diagrams (DFD), including Context and Level 1 DFDs, with external entities, processes, data stores, and flows.

Requirements: Addressed non-functional requirements like scalability and reliability. *Format:* Formatted in LaTeX, converted to Word, approximately 2,000 words.

Statistical Analysis Projects

- Computed multiplicative seasonal indices using least squares for Kotoka International Airport passenger traffic data (2022-2023).
Process: Deseasonalized data, calculated centered moving averages, adjusted indices to sum to 1200%.
Outcome: Fitted linear trends for forecasting.
- Applied the strong law of large numbers and central limit theorem to determine die tosses for relative frequency approximation (at least 5,337 tosses for 95% confidence within 0.01 of $1/6$).
- Solved vector problems involving components, projections (scalar/vector), perpendicularity, centroids, and geometric proofs.

Other Projects

- Redesigned status indicators for color-blind accessibility, including mockups/sketches.
- Created educational flyers on topics like mineral salts (functions, deficiencies, prevention, health benefits).

Interests and Extracurriculars

Passionate about integrating computer science with statistics for data-driven solutions. Enjoy exploring mathematical proofs and real-world applications in optimization and modeling. Active in university projects and team collaborations.

References

Available upon request.